

ADAS Features: Technical Overview and Safety Requirements

ACC · AEB · LKA · BSD · TJA · Highway Pilot · Sensor Fusion

Knowledge Base Document · RAG-Ready · ADAS Domain

1. SAE Levels of Driving Automation

The Society of Automotive Engineers (SAE) J3016 standard defines six levels of driving automation, widely used as the framework for describing ADAS and autonomous driving capability:

Level	Name	Description	Driver Role	Examples
L0	No Automation	All dynamic driving tasks performed by human	Full control	Conventional vehicle
L1	Driver Assist	One aspect of driving assisted (steering or speed)	Monitoring & Ready to take over	ACC, LKA
L2	Partial Automation	Both steering and speed controlled simultaneously	Monitoring & Ready to take over	ACC, LKA, Tesla Autopilot (basic)
L3	Conditional Automation	System handles all driving in defined conditions	Monitoring & Ready to take over	Highway Pilot, Traffic Jam Assist
L4	High Automation	System handles all driving in all conditions	No driver required	Robotaxi in geo-fenced area
L5	Full Automation	System handles all driving in all conditions	No driver required	Full self-driving (not yet deployed at scale)

2. Adaptive Cruise Control (ACC)

ACC automatically adjusts vehicle speed to maintain a set speed and time gap to a preceding vehicle, using radar and/or camera sensors for target detection and tracking.

2.1 Functional Description

- **Speed control:** Maintains set speed when no lead vehicle is detected. Uses throttle and engine braking; some systems use mild friction braking.
- **Gap control:** When a lead vehicle is detected, maintains time gap (typically 1.0–2.5 seconds set by driver). Adjusts speed via throttle and braking as lead vehicle speed changes.
- **Target selection:** Multi-object tracking identifies the most relevant lead vehicle. Relevance scoring considers lane assignment, TTC, range, and trajectory prediction. Cut-in and cut-out handling are critical edge cases.
- **Stop-and-go:** Extended ACC can brake to a full stop and resume from standstill. Required for urban and traffic jam use cases (Traffic Jam Assist).

2.2 Sensor Architecture

- **Long-range radar (77 GHz):** Primary sensor. Range up to 250 m, narrow FOV (~±10°). Provides range, range rate (Doppler velocity), and azimuth angle. Robust in adverse weather.
- **Front camera:** Provides object classification (car, truck, motorcycle, pedestrian), lane assignment, and traffic sign reading for speed limit adaptation. Fused with radar.
- **Medium-range radar:** Some systems add 60°-wide medium-range radar for cut-in detection in adjacent lanes.

2.3 ACC Safety Requirements (ISO 26262)

- **ASIL B – Unintended deceleration:** ACC shall not cause unintended deceleration exceeding defined thresholds (typically 3.5 m/s²) without driver intent. Safety mechanism: plausibility check of braking command against driver input and road context.
- **ASIL C – False target tracking:** ACC shall not track infrastructure objects (bridges, tunnels, gantries) as lead vehicles and apply false braking. Safety mechanism: object classification filter, height plausibility, stationary target suppression above speed threshold.
- **ASIL B – Target loss handling:** Upon loss of lead vehicle tracking, ACC shall not immediately accelerate to set speed if a vehicle may still be present. Time-delayed acceleration with driver notification.
- **Safe state:** ACC deactivation with audio/visual HMI alert. Driver takes over speed control.

2.4 ACC Performance KPIs

KPI	Target	Measurement Method
Target detection range	≥150 m at highway speeds	Validation test runs
Target selection accuracy	<2% missed target rate (cut-in scenarios)	ADT56 simulator on scenario suite
False target rate	<0.1/hour of operation	Long-duration fleet testing
Brake response time	<200 ms from target detection to brake initiation	Hardware-in-loop testing
Speed control accuracy	±2 km/h of set speed	Track testing
Stop-and-go: resume delay	<0.5 s after lead vehicle starts moving	Standardised test protocol

3. Automatic Emergency Braking (AEB)

AEB automatically applies emergency braking to prevent or mitigate collisions when the driver fails to respond to an imminent collision threat. AEB is increasingly mandated by regulation (Euro NCAP, NHTSA, EC Reg 2019/2144) and typically carries ASIL C–D requirements.

3.1 AEB Variants

- **Forward Collision Warning (FCW):** Auditory/visual alert when $TTC < \text{threshold}$. No braking. L0 function, but feeds into AEB timing.
- **AEB City:** Low-speed (0–60 km/h) operation. Targets pedestrians, cyclists, and vehicles. Typically ASIL C–D.
- **AEB Highway:** High-speed (>60 km/h) operation. Primarily vehicle targets. ASIL D for rear-end collision scenarios.
- **AEB Cyclist / AEB Pedestrian:** Specific detection pipelines for vulnerable road users. Lower confidence thresholds due to importance, higher false activation risk.
- **AEB Reverse:** Rear cross-traffic braking during reversing manoeuvres.
- **AEB Junction Assist:** Detects cross-traffic at junctions. Complex due to high target variety.

3.2 AEB Architecture

A typical AEB pipeline consists of:

- **1. Threat assessment:** Computes TTC (Time-to-Collision) and TTS (Time-to-Steer). Multiple algorithms run in parallel: kinematic TTC, prediction-based TTC, and occupancy-based.
- **2. Object confirmation:** Multi-sensor confirmation (radar + camera) required before full braking. Camera provides object classification and lateral position; radar provides precise range and velocity. A threat confirmed by both sensors triggers full braking.
- **3. Brake request generation:** Deceleration request computed based on required deceleration to avoid collision. Sent to brake controller via ISO 26262-protected interface (E2E).
- **4. Independent safety monitor:** Separate processing path validates plausibility of brake request before actuation. Checks: speed plausibility, road type (not parking lot), driver override detection.

3.3 AEB Safety Architecture

- **ASIL D – False activation prevention:** AEB shall not apply emergency braking in situations where no collision hazard exists. Primary mechanism: dual-channel confirmation (radar AND camera). Secondary: independent safety monitor SW component, E2E protected brake request.
- **ASIL D – Missed activation prevention:** AEB shall activate emergency braking for genuine frontal collision threats. Primary mechanism: redundant threat assessment algorithms. Secondary: minimum braking requirement (partial braking even if full confirmation not achieved).
- **Infrastructure suppression:** AEB must reliably suppress false activations for overhead structures, road signs, and parked vehicles on adjacent lanes. This is one of the hardest engineering problems in AEB — overly aggressive suppression reduces genuine hazard coverage.

4. Lane Keeping Assist (LKA) and Lane Departure Warning (LDW)

4.1 LDW – Lane Departure Warning

LDW monitors lane markings and alerts the driver when the vehicle crosses a lane boundary without an active turn signal. The warning is typically auditory (rumble strip sound), haptic (steering wheel vibration), or visual.

- **Sensor:** Front-facing camera. ASIL A–B (warning only, no actuation).
- **Lane marking detection:** Computer vision pipeline detects white/yellow lane markings using edge detection, Hough transform or learned CNN models. Lane model fitted as clothoid curves.
- **Departure prediction:** Time-to-lane-crossing (TTLC) computed from lateral velocity and distance to marking. Warning triggered at $TTLC < 0.5\text{--}1.0\text{ s}$.

4.2 LKA – Lane Keeping Assist

LKA actively applies steering torque to keep the vehicle within lane boundaries. It supplements driver steering rather than replacing it — maximum torque typically 3–5 Nm.

- **ASIL B–C:** Unintended steering actuation or failure to release control to driver requires ASIL B–C safety architecture.
- **Driver override:** LKA must immediately release upon driver steering torque input exceeding threshold (typically 2 Nm). Safety mechanism: steering torque sensor monitoring.
- **Turn signal inhibit:** LKA deactivates on turn signal activation to permit intentional lane changes. Failure to deactivate is a safety violation.
- **Lane confidence:** LKA must deactivate when lane marking confidence falls below threshold (worn markings, construction zones, snow cover). Confidence estimate from vision system is a safety-relevant input requiring E2E protection or plausibility monitoring.

5. Sensor Fusion in ADAS

Modern ADAS systems use multiple sensor modalities to achieve the detection performance, redundancy, and robustness required by safety requirements. Each sensor type has complementary strengths and weaknesses.

Sensor	Strengths	Weaknesses	Primary ADAS Use
77 GHz Radar	All weather, precise velocity (Doppler), long range (250 m)	Poor angular resolution, no object classification	ACC gap control, AEB primary threat sensor
Camera	Object classification, lane detection, sign reading	Degraded in rain/fog/night, no direct velocity measurement	LKA, LDW, sign recognition, AEB object confirmation
LiDAR	3D point cloud, precise geometry, good classification	Expensive, degraded in rain, no Doppler velocity	L3+ perception, HD mapping, free space detection
Ultrasonic	Short range (3 m), robust, low cost	Very short range only, sensitive to dirt	Parking assist, low-speed AEB reverse
Infrared/Thermal	Pedestrian detection at night, long range	No colour/texture, high cost, export restrictions	Night vision, pedestrian warning (premium vehicles)

5.1 Radar-Camera Fusion

The most common fusion configuration in production ADAS combines long-range radar with a front camera. Fusion is typically performed at the object list level (late fusion):

- **Track-to-track association:** Radar tracks and camera detections are associated using gating (spatial proximity) followed by probabilistic assignment (Hungarian algorithm or JPDA). Associated tracks are fused using Kalman filter state estimation.
- **Complementary attributes:** Fused object track combines radar's precise range/velocity with camera's classification and lateral position. The fused object carries both an existence probability and a classification confidence.
- **Conflict detection:** Significant disagreement between radar range and camera-estimated range triggers a plausibility fault — a key safety mechanism for AEB false activation prevention.

5.2 Multi-Object Tracking

Multi-object tracking (MOT) maintains persistent tracks of objects across sensor frames. Each track has a state estimate (position, velocity, acceleration, heading) and a track quality metric. Key algorithms:

- **Kalman Filter:** Optimal state estimator for linear systems with Gaussian noise. Extended Kalman Filter (EKF) or Unscented Kalman Filter (UKF) for non-linear motion models.
- **Track management:** Track initialisation (tentative → confirmed after N consecutive detections), track deletion (confirmed → deleted after M consecutive misses), and track merging for duplicates.
- **JPDA / MHT:** Joint Probabilistic Data Association and Multiple Hypothesis Tracking handle ambiguous sensor-to-track associations in dense traffic scenarios.

ADAS technical overview. RAG pipeline use. KPIs and safety levels are representative industry values; actual product requirements vary by OEM, vehicle programme, and regulation.